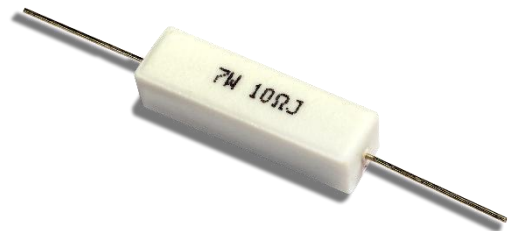


# Ceramic Case Resistors - Wirewound / Metal Oxide

## SQP Series

### Features:

- Ratings from 2W to 25W
- Resistance from 0R1 to 200K
- High overload capability
- Flameproof ceramic case
- Inorganic potting material



All parts are Pb-free and comply with EU Directive 2011/65/EU amended by (EU) 2015/863

## Electrical Data

|                                    | SQP2                 | SQP3     | SQP5     | SQP7S     | SQP7      | SQP10    | SQP15    | SQP20    | -     |
|------------------------------------|----------------------|----------|----------|-----------|-----------|----------|----------|----------|-------|
| Legacy type - wirewound            | CAW2                 | CAW3     | CAW5     | -         | CAW7      | CAW10    | CAW15    | CAW20    | CAW25 |
| Legacy type – oxide film           | CAF2                 | CAF3     | CAF5     | -         | CAF7      | CAF10    | CAF15    | CAF20    | -     |
| Power rating @70°C W               | 2                    | 3        | 5        | 7         |           | 10       | 15       | 20       | 25    |
| Limiting element voltage V         | 250                  | 300      | 350      | 500       | 700       | 750      |          |          |       |
| Resistance range - wirewound ohms  | 0R1–27R              | 0R1–39R  | 0R1–47R  | 0R1–100R  | 0R1–680R  | 0R1–910R | 1R0–1K0  | 2R0–1K2  |       |
| Resistance range – oxide film ohms | 30R–120K             | 43R–150K | 51R–150K | 110R–150K | 750R–200K | 1K0–200K | 1K1–200K | 1K3–200K | -     |
| Resistance tolerance %             | 5                    |          |          |           |           |          |          |          |       |
| TCR ppm/°C                         | <20R: 400, ≥20R: 350 |          |          |           |           |          |          |          |       |
| Isolation voltage V                | 1000                 |          |          |           |           |          |          |          |       |
| Standard values                    | E24 preferred        |          |          |           |           |          |          |          |       |
| Thermal impedance °C/W             | 50                   | 45       | 30       | 26        | 21        | 16       | 12       | 10       |       |
| Ambient temperature range °C       | -55 to 155           |          |          |           |           |          |          |          |       |

## Physical Data

| Dimensions in mm and weight in g |       |       |       |       |        |         |  |
|----------------------------------|-------|-------|-------|-------|--------|---------|--|
| Type                             | L max | W max | H max | f min | d ±0.1 | Wt. nom |  |
| SQP2                             | 19    | 8     | 23    | 0.7   | 2.2    |         |  |
| SQP3                             | 23    | 9     | 27    | 0.7   | 3.5    |         |  |
| SQP5                             | 23    | 11    | 10    | 30    | 0.75   | 5       |  |
| SQP7S                            | 26    | 10    | 10    | 30    | 0.75   | 6       |  |
| SQP7                             | 36    | 11    | 10    | 30    | 0.75   | 8.2     |  |
| SQP10                            | 50    | 11    | 10    | 30    | 0.75   | 11      |  |
| SQP15                            | 50    | 13.5  | 12.5  | 30    | 0.75   | 17.4    |  |
| SQP20                            | 61    | 15.5  | 14.5  | 30    | 0.75   | 25      |  |
| CAW25                            | 65    | 15.5  | 14.5  | 30    | 0.75   | 26.7    |  |

### Construction

SQP resistors have a high purity ceramic rod, with force fit end caps. Depending on value, the element is either wirewound or a deposited metal oxide film. This has termination wires welded to the caps and is fitted into a ceramic case with fireproof insulation cement.

### Marking

Power rating, resistance value and tolerance code are legend marked onto the upper surface.

### Terminations

**Material** Tinned copper.

**Strength** The terminations meet the requirements of IEC 60115-1 clause 9.5.

**Solderability** The terminations meet the requirements of IEC 60115-1 clause 11.1.

### Flammability

The resistor will not burn or emit incandescent particles under any condition of applied temperature or overload.

### General Note

TT Electronics reserves the right to make changes in product specification without notice or liability.  
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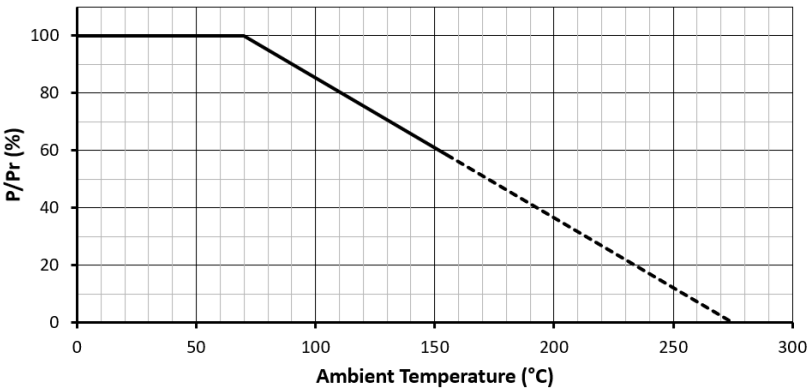
Solvent resistance

The body protection and marking are resistant to all normal industrial solvents suitable for printed circuits.

Performance Data

|                                                                      |      | Maximum (+R05)            |
|----------------------------------------------------------------------|------|---------------------------|
| Load at rated power: P <sub>r</sub> for 1000 hours at 70°C           | ±ΔR% | ≤100K: 5, >100K: 10       |
| Short term overload: Lesser of 6.25xP <sub>r</sub> or 2.5xLEV for 5s | ±ΔR% | 2                         |
| Damp heat steady state: 56 days, 40°C, ≥90% RH                       | ±ΔR% | 2                         |
| Temperature rapid change: 5 cycles, -55/155°C                        | ±ΔR% | 2                         |
| Resistance to solder heat                                            | ±ΔR% | 1                         |
| Pulse handling                                                       |      | Data available by request |

Temperature Derating



Ordering Procedure

Global Part Number Example: SQP3-1K2JB5 (SQP3, 1.2 kilohms ±5%, Pb-free)

|   |   |   |   |  |   |   |   |   |  |   |   |   |  |
|---|---|---|---|--|---|---|---|---|--|---|---|---|--|
| S | Q | P | 3 |  | - | 1 | K | 2 |  | J | B | 5 |  |
| 1 |   |   |   |  |   | 2 |   |   |  | 3 | 4 |   |  |

| 1<br>Type | 2<br>Value                                      | 3<br>Tolerance | 4<br>Packing |                |
|-----------|-------------------------------------------------|----------------|--------------|----------------|
| SQP2      | E24 = 3/4 characters<br>R = ohms<br>K = kilohms | J = ±5%        | B5           | SQP2 3600/box  |
| SQP3      |                                                 |                |              | SQP3 3000/box  |
| SQP5      |                                                 |                |              | SQP5 2400/box  |
| SQP7S     |                                                 |                | B15          | SQP7S 1800/box |
| SQP7      |                                                 |                |              | SQP7 1500/box  |
| SQP10     |                                                 |                |              | SQP10 840/box  |
| SQP15     |                                                 |                | B1           | SQP15 720/box  |
| SQP20     |                                                 |                |              | SQP20 720/box  |

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Legacy Part Numbers

This product has a legacy part number format. This is still available for ordering, but for new designs use of the Global Part Number is recommended. However, CAW25 is only available under the legacy part number.

Legacy Part Number Example: CAF31201JLF (CAF3, 1.2 kilohms ±5%, Pb-free)



| 1<br>Type    | 2<br>Body Shape                                            | 3<br>Value                                                | 4<br>Tolerance | 5<br>Termination | Packing |          |
|--------------|------------------------------------------------------------|-----------------------------------------------------------|----------------|------------------|---------|----------|
| CAW2, CAF2   | Omit for standard<br>A = Alternate<br>(2, 5, 7 & 10W only) | 3 digits + multiplier<br>R = ohms for values<br><100 ohms | J = ±5%        | LF = Pb-free     | CA-2    | 3600/box |
| CAW3, CAF3   |                                                            |                                                           |                |                  | CA-3    | 3000/box |
| CAW5, CAF5   |                                                            |                                                           |                |                  | CA-5    | 2400/box |
| CAW7, CAF7   |                                                            |                                                           |                |                  | CA-7    | 1800/box |
| CAW10, CAF10 |                                                            |                                                           |                |                  | CA-10   | 1500/box |
| CAW15, CAF15 |                                                            |                                                           |                |                  | CA-15   | 840/box  |
| CAW20, CAF20 |                                                            |                                                           |                |                  | CA-20   | 720/box  |
| CAW25        |                                                            |                                                           |                |                  |         |          |